

R for SAS Users

Melinda Higgins

<https://statisticalhorizons.com/seminars/r-sas-users/>

2-day Workshop

- Day 1:
 - 10:30 - 12:30 am
 - 30 minute break
 - 1:00 – 3:00 pm
- Day 2:
 - 10:30 - 12:30 am
 - 30 minute break
 - 1:00 - 3:00 pm

Workshop Resources

- GoogleDrive
 - <https://drive.google.com/drive/folders/1Nnb9pA7g7iRLg9o2yidV2MWMWoDT0cHV?usp=sharing>
- WEBSITE:
 - <https://tinyurl.com/r4sasusers-oct2026>
- SLACK:
 - <https://r-for-sas-users-sh.slack.com>
- H.E.L.P. Dataset and “SAS and R” Book
 - <https://nhorton.people.amherst.edu/sasr2/datasets.php>
 - <https://nhorton.people.amherst.edu/sasr2/>

Compare R and SAS

R

- Bare Bones (89MB)
- Takes up little memory
- Powerful base packages included
- FREE/Open Source
- Add “packages” as you go – only “load” what you need at that time

SAS

- Have to buy base \$\$ & Large (20GB+)
- Almost always have to buy add-ons, gets expensive \$\$
- Must know what you want ahead of time
- Sometimes end up with more than you need

My Current SAS License Includes:

```
1 sascode_d1_01.sas
NO
* =====;
* get details on SAS components and
  products installed;
* =====;
proc setinit; run;
```

Warning Period: 15 days (ending 29AUG2026).
System birthday: 08JUL2025.
Operating System: WX64_WKS.
Product expiration dates:

---Base SAS Software	14AUG2026 (CPU A)	---SAS Bridge for ESRI	14AUG2026 (CPU A)	---SAS/ACCESS Interface to Netezza	14AUG2026 (CPU A)
---SAS/STAT	14AUG2026 (CPU A)	---OR OPT	14AUG2026 (CPU A)	---SAS/ACCESS Interface to Greenplum	14AUG2026 (CPU A)
---SAS/GRAPH	14AUG2026 (CPU A)	---OR PRS	14AUG2026 (CPU A)	---SAS/ACCESS Interface to SAP IQ	14AUG2026 (CPU A)
---SAS/ETS	14AUG2026 (CPU A)	---OR IVS	14AUG2026 (CPU A)	---SAS/ACCESS to Hadoop	14AUG2026 (CPU A)
---SAS/FSP	14AUG2026 (CPU A)	---OR LSO	14AUG2026 (CPU A)	---SAS/ACCESS to Vertica	14AUG2026 (CPU A)
---SAS/OR	14AUG2026 (CPU A)	---SAS/ACCESS Interface to DB2	14AUG2026 (CPU A)	---SAS/ACCESS to Postgres	14AUG2026 (CPU A)
---SAS/AF	14AUG2026 (CPU A)	---SAS/ACCESS Interface to Oracle	14AUG2026 (CPU A)	---SAS/ACCESS to Impala	14AUG2026 (CPU A)
---SAS/IML	14AUG2026 (CPU A)	---SAS/ACCESS Interface to SAP ASE	14AUG2026 (CPU A)	---SAS/ACCESS to Salesforce	14AUG2026 (CPU A)
---SAS/QC	14AUG2026 (CPU A)	---SAS/ACCESS Interface to PC Files	14AUG2026 (CPU A)	---SAS/ACCESS to Amazon Redshift	14AUG2026 (CPU A)
---SAS/SHARE	14AUG2026 (CPU A)	---SAS/ACCESS Interface to ODBC	14AUG2026 (CPU A)	---High Performance Suite	14AUG2026 (CPU A)
---SAS/ASSIST	14AUG2026 (CPU A)	---SAS/ACCESS Interface to OLE DB	14AUG2026 (CPU A)	---SAS Add-in for Microsoft Excel	14AUG2026 (CPU A)
---SAS/CONNECT	14AUG2026 (CPU A)	---SAS/ACCESS Interface to R/3	14AUG2026 (CPU A)	---SAS/ACCESS to SAP HANA	14AUG2026 (CPU A)
---SAS/EIS	14AUG2026 (CPU A)	---SAS/ACCESS Interface to Teradata	14AUG2026 (CPU A)	---PRODNUM0896	14AUG2026 (CPU A)
---SAS/GIS	14AUG2026 (CPU A)	---SAS/ACCESS Interface to Microsoft SQL Server	14AUG2026 (CPU A)	---SAS/ACCESS Interface to the PI System	14AUG2026 (CPU A)
---SAS/SHARE*NET	14AUG2026 (CPU A)	---SAS/ACCESS Interface to MySQL	14AUG2026 (CPU A)	---SAS/ACCESS to JDBC	14AUG2026 (CPU A)
---SAS Enterprise Miner		---SAS Enterprise Miner for Desktop	14AUG2026 (CPU A)	---prod 1312	14AUG2026 (CPU A)
		---SAS/IML Studio	14AUG2026 (CPU A)		
		---SAS Workspace Server for Local Access			

SAS Software Components

```
2 proc product_status; run;
```

For Base SAS Software ...
Custom version information: 9.4_M6
Image version information: 9.04.01M6P110718

* For SAS/STAT ...
Custom version information: 15.1

* For SAS/GRAPH ...
Custom version information: 9.4_M6

* For SAS/ETS ...
Custom version information: 15.1

* For SAS/FSP ...
Custom version information: 9.4_M6

* For SAS/OR ...
Custom version information: 15.1
Image version information: 9.04.01M6P050819

* For SAS/AF ...
Custom version information: 9.4_M6

* For SAS/IML ...
Custom version information: 15.1

* For SAS/QC ...
Custom version information: 15.1

* For SAS/SHARE ...
Custom version information: 9.4_M5

* For SAS/ASSIST ...
Custom version information: 9.4
Image version information: 9.04.01M0P061913

* For SAS/CONNECT ...
Custom version information: 9.4_M6

* For SAS/ACCESS to JDBC ...
Custom version information: 1.0

* For SAS/EIS ...
Custom version information: 9.4_M6

* For SAS/GIS ...
Custom version information: 9.4_M6

* For SAS/ACCESS Interface to Netezza ...
Custom version information: 9.43

* For SAS/ACCESS Interface to Greenplum ...
Custom version information: 9.43

* For SAS/ACCESS Interface to SAP IQ ...
Custom version information: 9.4_M5

* For SAS/ACCESS to Hadoop ...
Custom version information: 9.46

* For SAS Enterprise Miner ...
Custom version information: 15.1

* For SAS/ACCESS to Vertica ...
Custom version information: 9.4_M6

* For SAS/ACCESS to Postgres ...
Custom version information: 9.4_M6

=====;

product_status; long write; s. It whether use;

=====

For SAS/ACCESS to Impala ...
Custom version information: 9.45

For SAS/ACCESS to Salesforce ...
Custom version information: 9.4
Image version information: 9.04.01M6P041019

For SAS/ACCESS to Amazon Redshift ...
Custom version information: 9.43

For SAS Integration Technologies ...
Custom version information: 9.4_M6

For SAS Text Miner ...
Custom version information: 15.1

For High Performance Suite ...
Custom version information: 2.2_M7

For SAS/ACCESS Interface to DB2 ...
Custom version information: 9.4_M2

For SAS/ACCESS Interface to Oracle ...
Custom version information: 9.43

For SAS/ACCESS Interface to the PI System ...
Custom version information: 9.44

For SAS/ACCESS Interface to SAP ASE ...
Custom version information: 9.4_M5

For SAS/ACCESS Interface to PC Files ...
Custom version information: 9.4_M6

For SAS/ACCESS Interface to ODBC ...
Custom version information: 9.4_M6

For SAS/ACCESS Interface to OLE DB ...
Custom version information: 9.4_M6

For SAS/ACCESS Interface to Teradata ...
Custom version information: 9.46

For SAS/ACCESS Interface to MySQL ...
Custom version information: 9.4_M6

NOTE: PROCEDURE PRODUCT_STATUS used (Total process time):
real time 0.12 seconds
cpu time 0.11 seconds

My Current R - Base Installation

```
Console Terminal x Background Jobs x
R 4.6.0 · ~/
> sessionInfo()
R version 4.6.0 (2026-04-24 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 22000)

Matrix products: default
LAPACK version 3.12.1

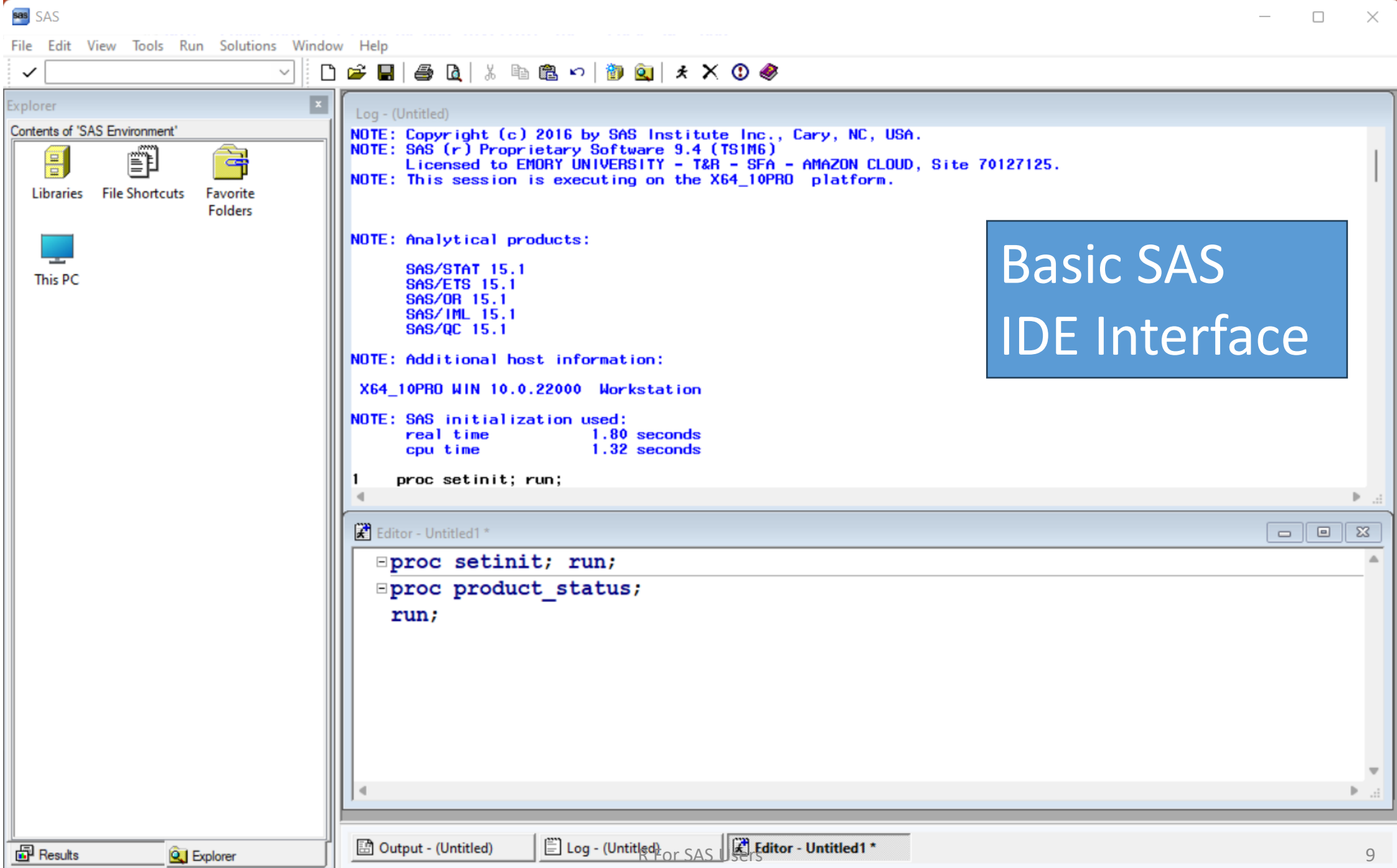
locale:
[1] LC_COLLATE=English_United States.utf8
[2] LC_CTYPE=English_United States.utf8
[3] LC_MONETARY=English_United States.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=English_United States.utf8

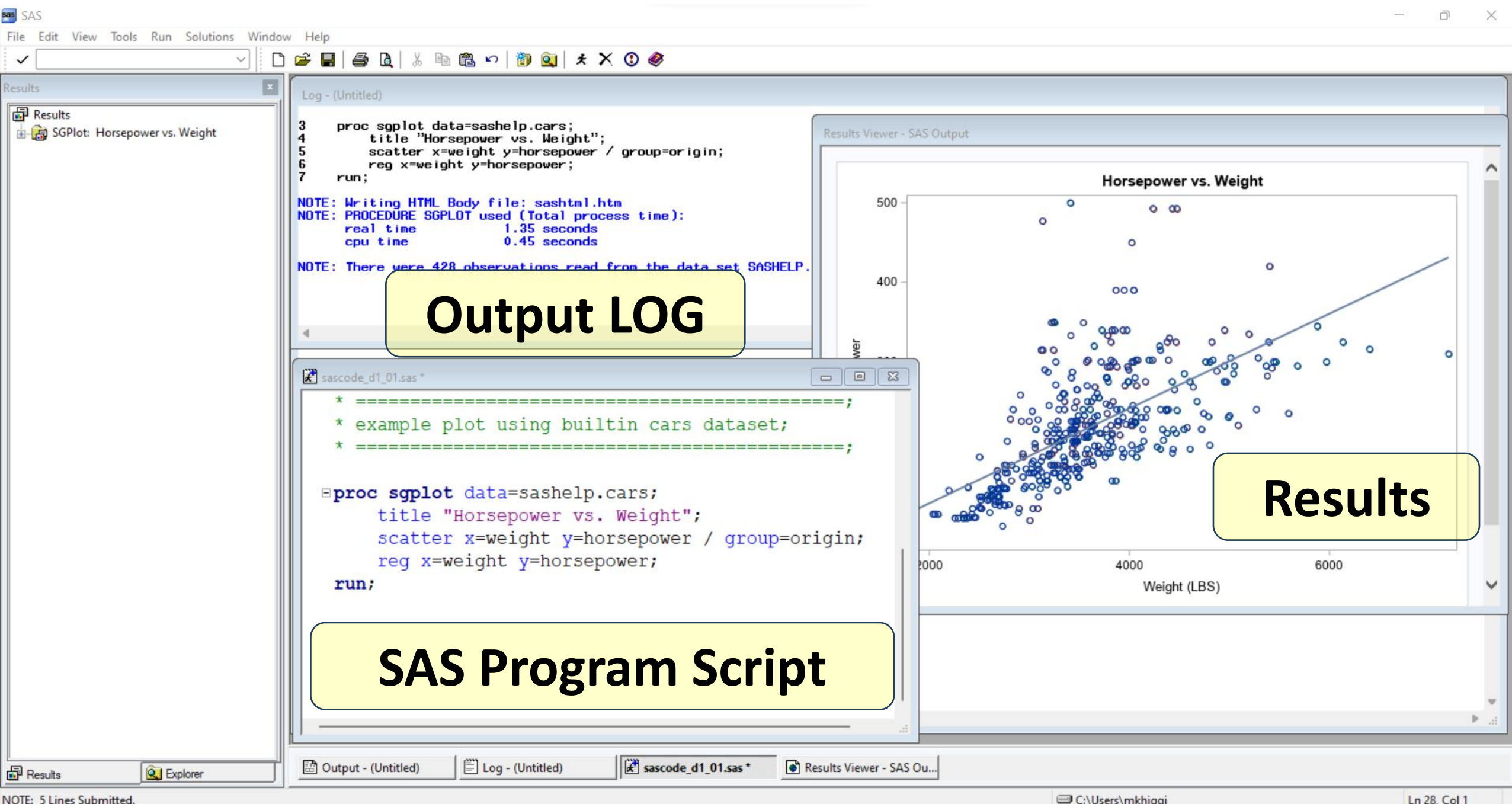
time zone: America/New_York
tzcode source: internal

attached base packages:
[1] stats      graphics  grDevices  utils      datasets  methods
[7] base
```

Only 7 Base
Packages

Interfaces – SAS Base; SAS Studio





SAS Studio Interface IDE/GUI

localhost:63567/main?locale=en_US&zone=G

SAS® Studio

SAS Programmer Sign Out

Files and Folders

- about_files
- archived
- cleandemo
- docs
- images
- installR
- rstudiocustom
- rstudioide
- sascodes
 - sascode_d1_01.sas
- slides
- _extensions
- _site
 - additionalResources.qmd
 - bioconductor_phyloseq.png
 - bioconductor_pkg.png
 - cc_by_nc_nd.png
 - compare_rggobi_tourr.png
 - compare_tablepkg.png

Tasks and Utilities

Snippets

Libraries

File Shortcuts

sascode_d1_01.sas

CODE LOG RESULTS

```
1 * =====;
2 * get details on SAS components and
3 * products installed;
4 * =====;
5
6 proc setinit; run;
7
8 * =====;
9 * returns a list of the SAS Foundation products
10 * that are installed on your system, along with
11 * the version numbers of those products. It
12 * provides a quick method to determine whether
13 * a SAS product is available for your use;
14 * =====;
15
16 proc product_status; run;
17
18 * =====;
19 * example plot using builtin dataset;
20 * =====;
21
22 proc sgplot data=sashelp.cars;
23     title "Horsepower vs. Weight";
24     scatter x=weight y=horsepower / group=origin;
25     reg x=weight y=horsepower;
26 run;
27
28
```

Line #

SAS Program Script

C:/MyGithub/StatisticalHorizons_R4SASUsers/sascodes/sascode_d1_01.sas

Line 28, Column 1 UTF-8

Messages: 1 User: Higgins, Melinda K

R For SAS Users

SAS Studio Interface IDE/GUI

localhost:63567/main?locale=en_US&zone=G

SAS® Studio

SAS Programmer Sign Out

Files and Folders

- about_files
- archived
- cleandemo
- docs
- images
- installR
- rstudiocustom
- rstudioide
- sascodes
 - sascode_d1_01.sas
- slides
- _extensions
- _site
 - additionalResources.qmd
 - bioconductor_phyloseq.png
 - bioconductor_pkg.png
 - cc_by_nc_nd.png
 - compare_rggobi_tourr.png
 - compare_tablepkg.png

Tasks and Utilities

Snippets

Libraries

File Shortcuts

sascode_d1_01.sas

CODE LOG RESULTS

Errors, Warnings, Notes

- Errors
- Warnings
- Notes (2)

Output LOG

```
1      OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
72
73      proc sgplot data=sashelp.cars;
74          title "Horsepower vs. Weight";
75          scatter x=weight y=horsepower / group=origin;
76          reg x=weight y=horsepower;
77      run;

NOTE: PROCEDURE SGPLOT used (Total process time):
      real time           2.98 seconds
      cpu time            1.12 seconds

NOTE: There were 428 observations read from the data set
      SASHELP.CARS.

78
79      OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
91
```

Messages: 1 User: Higgins, Melinda K

SAS Studio Interface IDE/GUI

← ↻ ⓘ localhost:63567/main?locale=en_US&zone=G A ☆ ☆ ... Chat

SAS® Studio SAS Programmer ? Sign Out

Files and Folders

- about_files
- archived
- cleandemo
- docs
- images
- installR
- rstudiocustom
- rstudioide
- sascode
 - sascode_d1_01.sas**
- slides
- _extensions
- _site
 - additionalResources.qmd
 - bioconductor_phyloseq.png
 - bioconductor_pkg.png
 - cc_by_nc_nd.png
 - compare_rggobi_tourr.png
 - compare_tablepkg.png

Tasks and Utilities

Snippets

Libraries

File Shortcuts

sascode_d1_01.sas

CODE LOG RESULTS

Table of Contents

Results

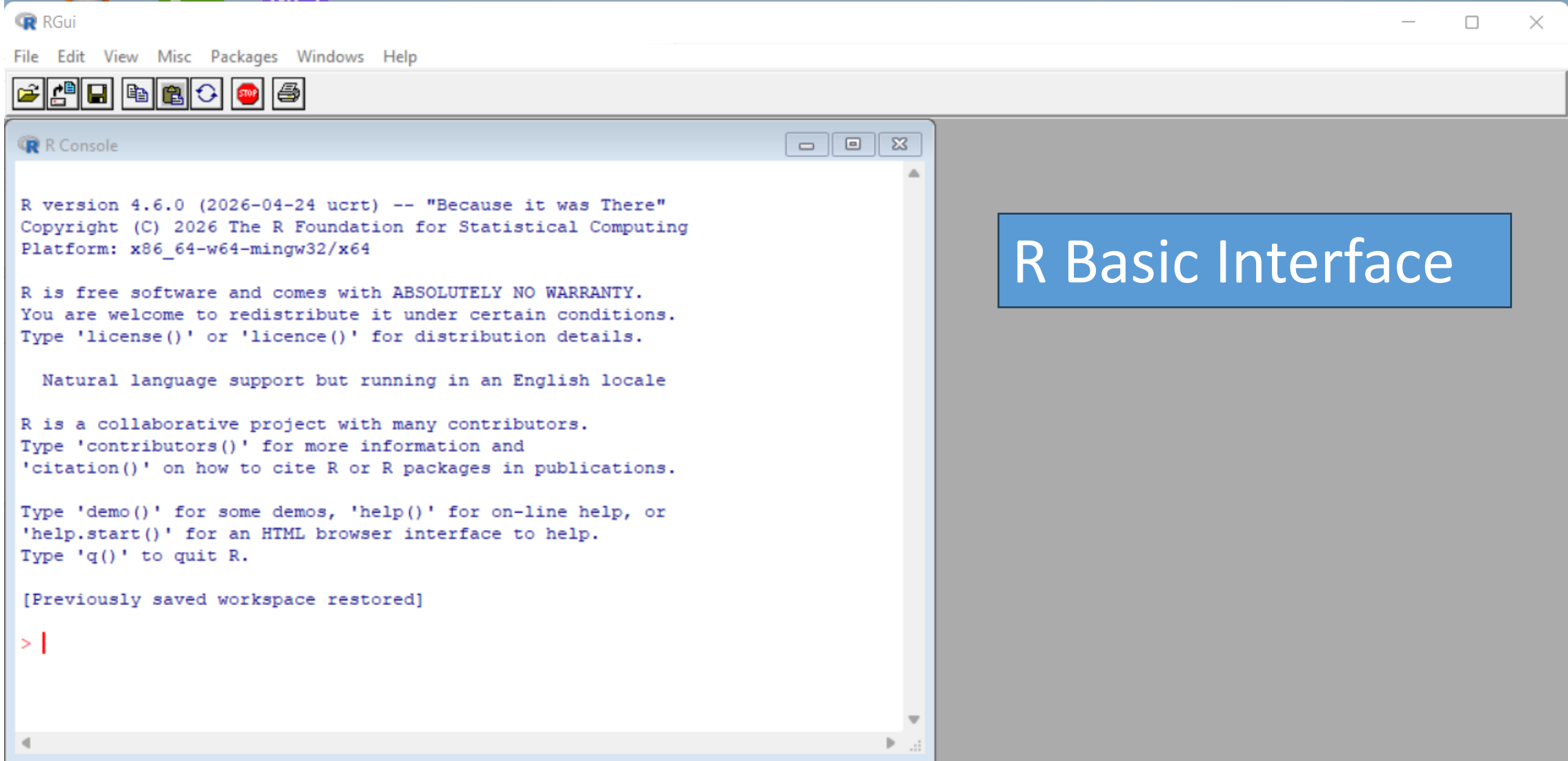
Horsepower vs. Weight

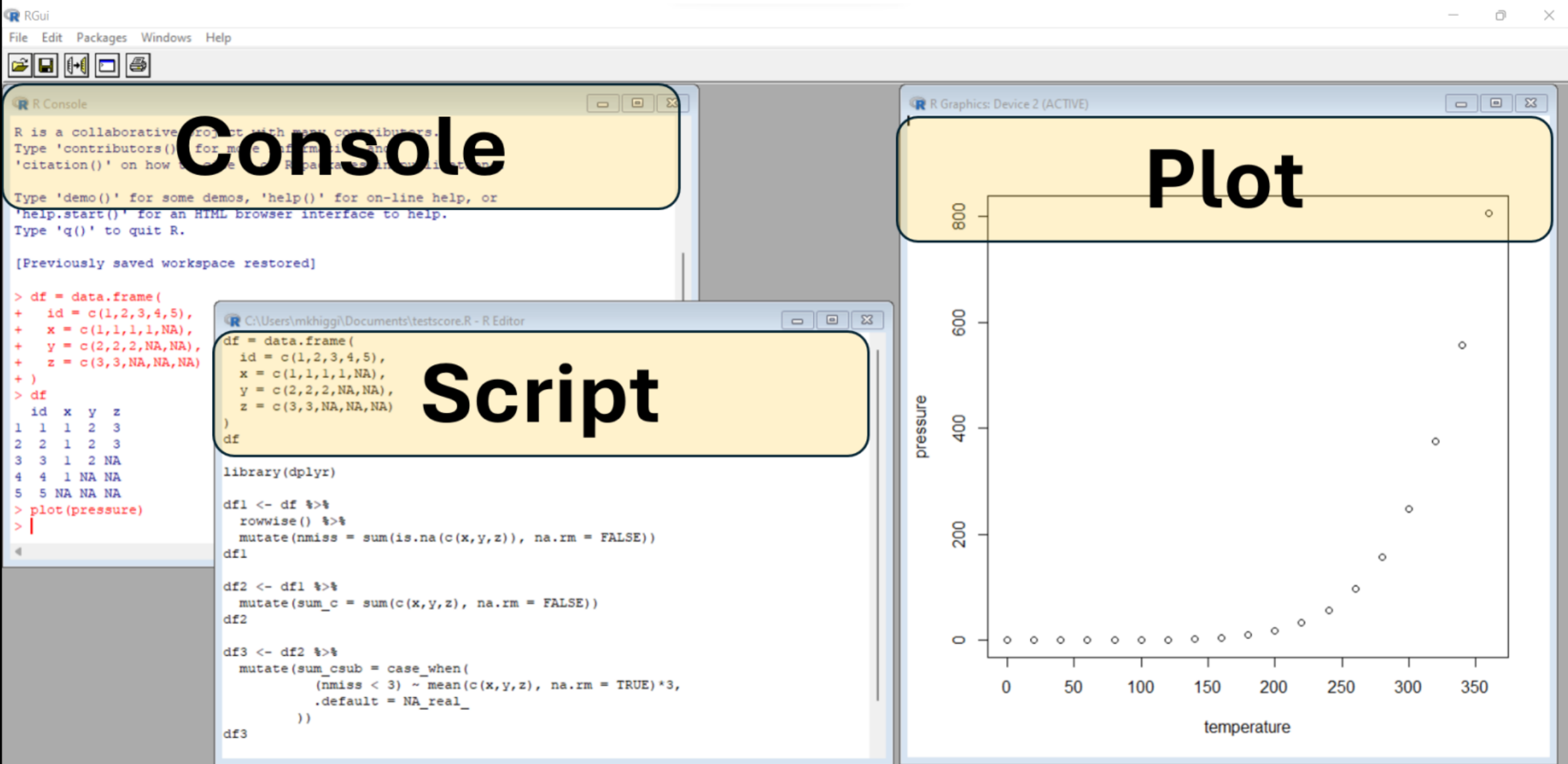
Origin: Asia Europe USA

Messages: 1 User: Higgins, Melinda K

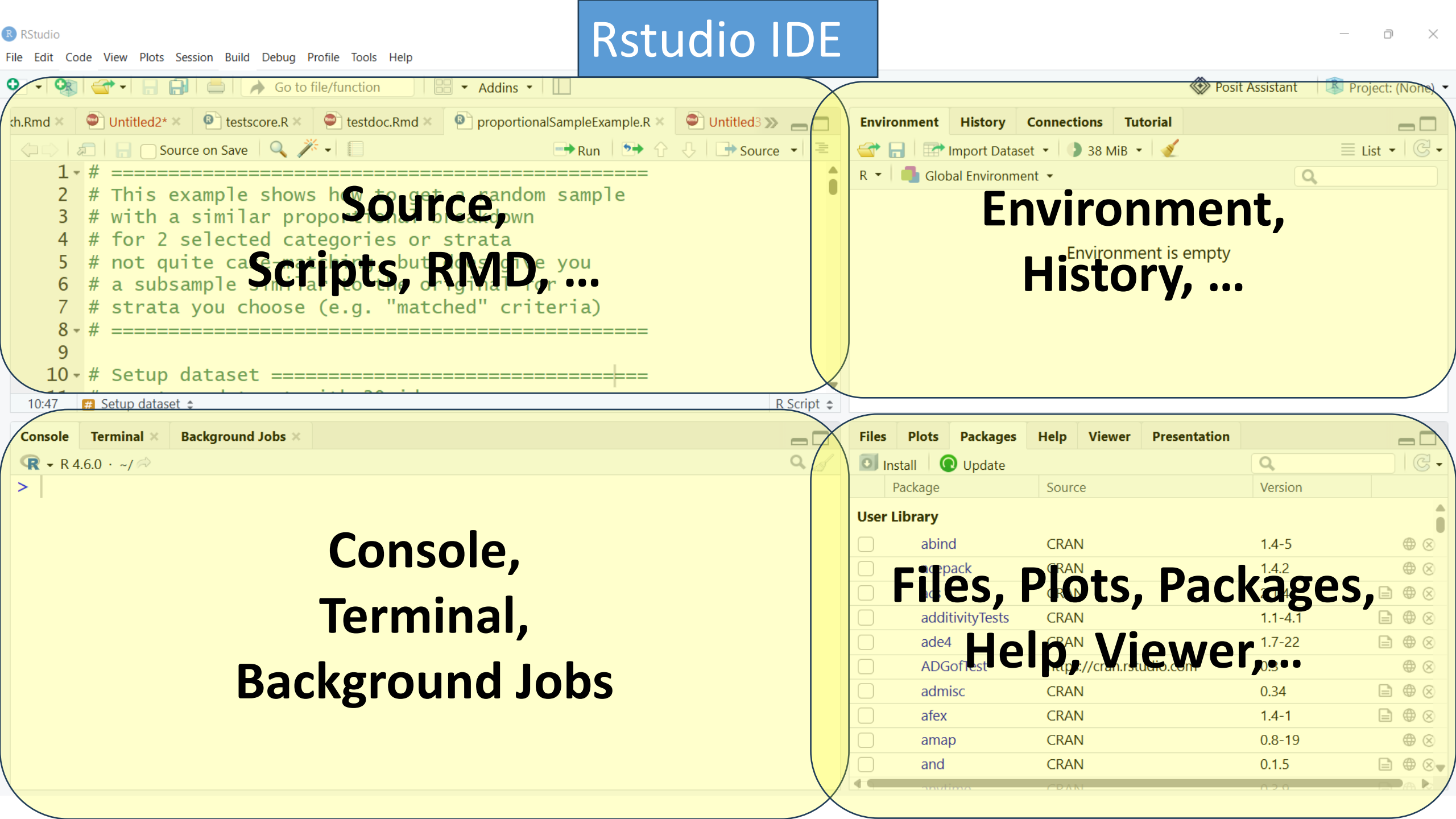
Interfaces – R Basic; Rstudio IDE; Positron

- R (from CRAN) – FREE – comes with a basic programming interface
 - <https://cran.r-project.org/>
- Rstudio is an IDE = Integrated Development Environment
 - <https://posit.co/download/rstudio-desktop>
- Positron is the next evolution of programming interfaces
 - <https://positron.posit.co/>
 - Learn more, <https://positron.posit.co/migrate-rstudio-compare.html>





R Basic Interface – similar to SAS Basic Interface



Rstudio IDE

**Source,
Scripts, RMD, ...**

**Environment,
History, ...**

**Console,
Terminal,
Background Jobs**

**Files, Plots, Packages,
Help, Viewer, ...**

Positron IDE

R 4.5.3

File Edit Selection View Go Run Terminal Help

New Open

EXPLORER

BSRPerceptions_07252025.qmd

OPEN EDITORS

BSRPerceptions_07252025.q...

NO FOLDER OPENED

You have not yet opened a folder.

Open Folder

Opening a folder will close all currently open editors. To keep them open, [add a folder](#) instead.

You can clone a repository locally.

Clone Repository

To learn more about how to use Git and source control in Positron, [read the VS Code docs](#).

OUTLINE

TIMELINE

C: > EMORY-SON > EHC_Projects_S.Shapiro_MGullatte > 2024_RobertaPolly_BesideShiftReport > B

The editor could not be opened because the file was not found.

Create File

CONSOLE

~ /

R 4.5.3 started.

R version 4.5.3 (2026-03-11 ucrt) -- "Reassured Reassurer"

Copyright (C) 2026 The R Foundation for Statistical Computing

Platform: x86_64-w64-mingw32/x64

R is free software and comes with ABSOLUTELY NO WARRANTY.

You are welcome to redistribute it under certain conditions.

Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.

Type 'contributors()' for more information and

'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or

'help.start()' for an HTML browser interface to help.

Type 'q()' to quit R.

SESSION

CONNECTIONS

HELP

HISTORY

VIEWER

VARIABLES

R 4.5.3

No variables have been created.

PLOTS

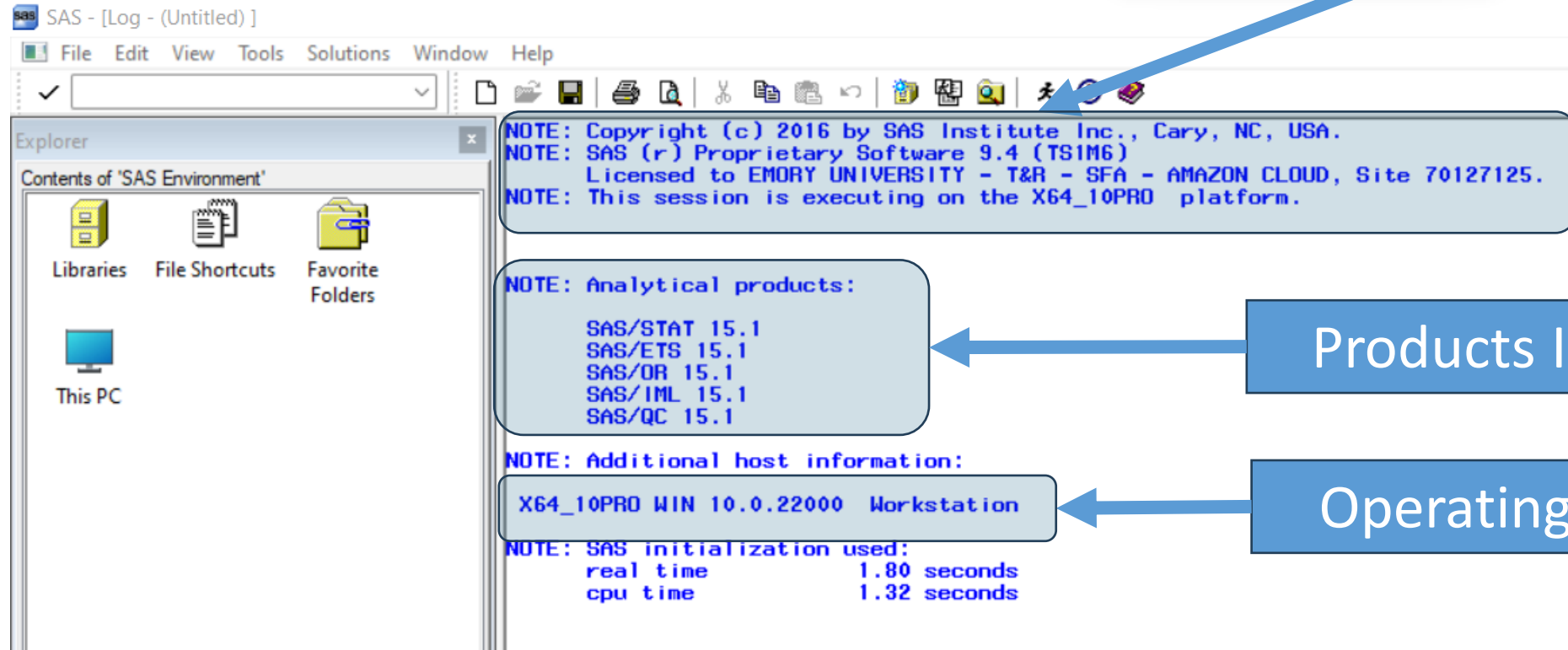
Quarto: 1.6.39

R For SAS Users

18

SAS Session

SAS Version and
Local System Details



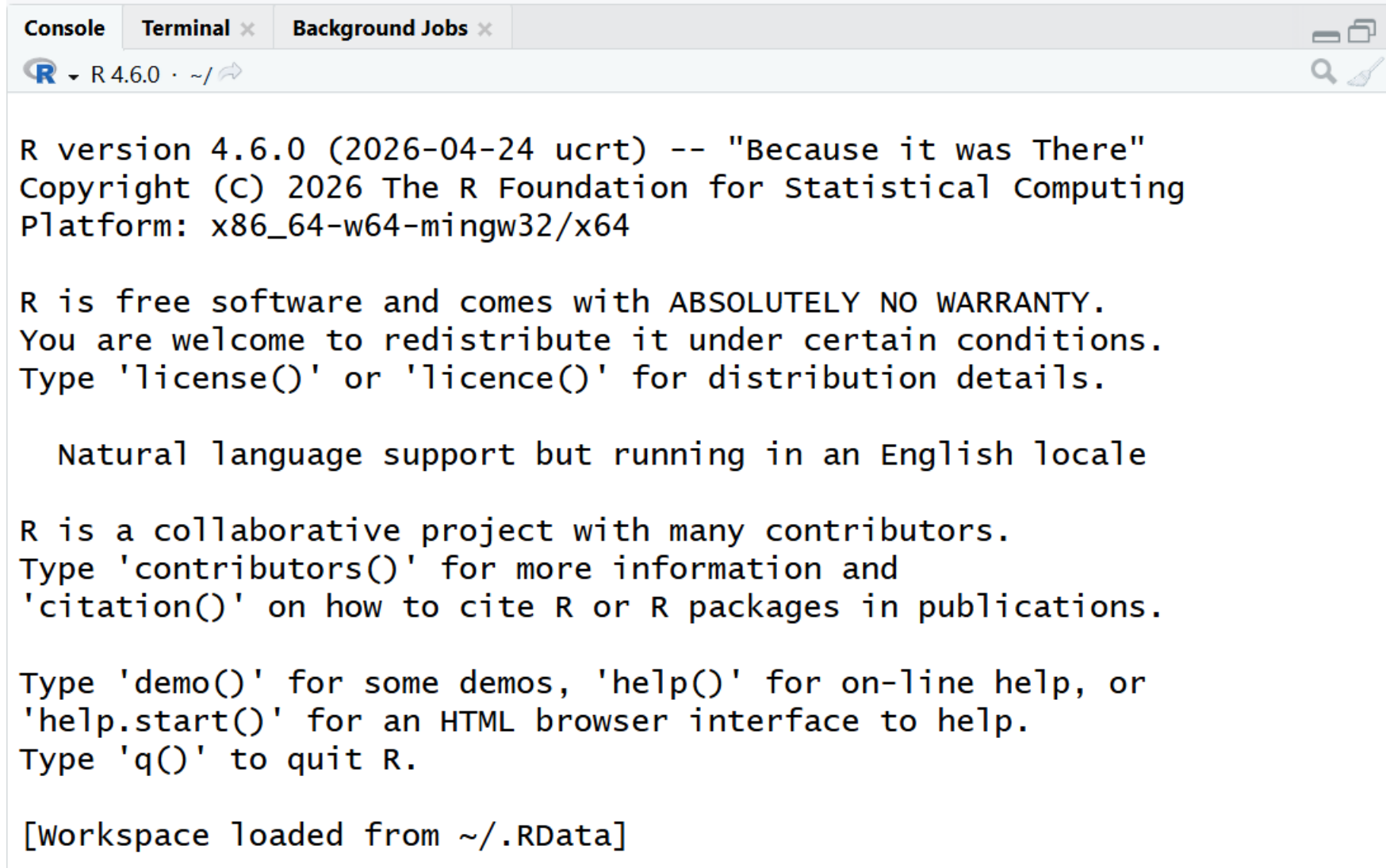
The screenshot shows the SAS Log window for a session titled 'SAS - [Log - (Untitled)]'. The log contains the following text:

```
NOTE: Copyright (c) 2016 by SAS Institute Inc., Cary, NC, USA.  
NOTE: SAS (r) Proprietary Software 9.4 (TS1M6)  
       Licensed to EMORY UNIVERSITY - T&R - SFA - AMAZON CLOUD, Site 70127125.  
NOTE: This session is executing on the X64_10PRO platform.  
  
NOTE: Analytical products:  
      SAS/STAT 15.1  
      SAS/ETS 15.1  
      SAS/OR 15.1  
      SAS/IML 15.1  
      SAS/QC 15.1  
  
NOTE: Additional host information:  
      X64_10PRO WIN 10.0.22000 Workstation  
  
NOTE: SAS initialization used:  
      real time      1.80 seconds  
      cpu time       1.32 seconds
```

Annotations with arrows point from the following text boxes to specific parts of the log:

- SAS Version and Local System Details** points to the first three lines of the log.
- Products Included** points to the 'Analytical products' section.
- Operating System** points to the 'Additional host information' section.

R Session – Details on Start



The screenshot shows an R console window with three tabs: 'Console', 'Terminal', and 'Background Jobs'. The 'Console' tab is active. The window title bar indicates 'R 4.6.0 · ~/'. The console output displays the R version (4.6.0), copyright (2026 The R Foundation for Statistical Computing), and platform (x86_64-w64-mingw32/x64). It also includes a disclaimer about the warranty and a list of useful functions like 'license()', 'contributors()', 'citation()', 'demo()', 'help()', 'help.start()', and 'q()'. The session ends with the message '[Workspace loaded from ~/.RData]'.

```
R version 4.6.0 (2026-04-24 ucrt) -- "Because it was There"
Copyright (C) 2026 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[Workspace loaded from ~/.RData]
```

R Session – sessionInfo()

The diagram illustrates the output of the `sessionInfo()` function in R, with callouts highlighting specific sections:

- R Version**: Points to the `> sessionInfo()` command and the first line of output, `R version 4.6.0 (2026-04-24 ucrt)`.
- Operating System**: Points to the `Platform` and `Running under` lines: `Platform: x86_64-w64-mingw32/x64` and `Running under: Windows 11 x64 (build 22000)`.
- Local Details**: Points to the `locale` and `time zone` sections:

```
locale:
[1] LC_COLLATE=English_United States.utf8
[2] LC_CTYPE=English_United States.utf8
[3] LC_MONETARY=English_United States.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=English_United States.utf8

time zone: America/New_York
tzcode source: internal
```
- 7 Base Packages**: Points to the `attached base packages` section:

```
attached base packages:
[1] stats      graphics  grDevices  utils      datasets  methods
[7] base
```

SAS

Components

+

WORK
Library

R

Packages

+

Global
Environment



SAS

```
PROC SETINIT  
PROC PRODUCT_STATUS
```



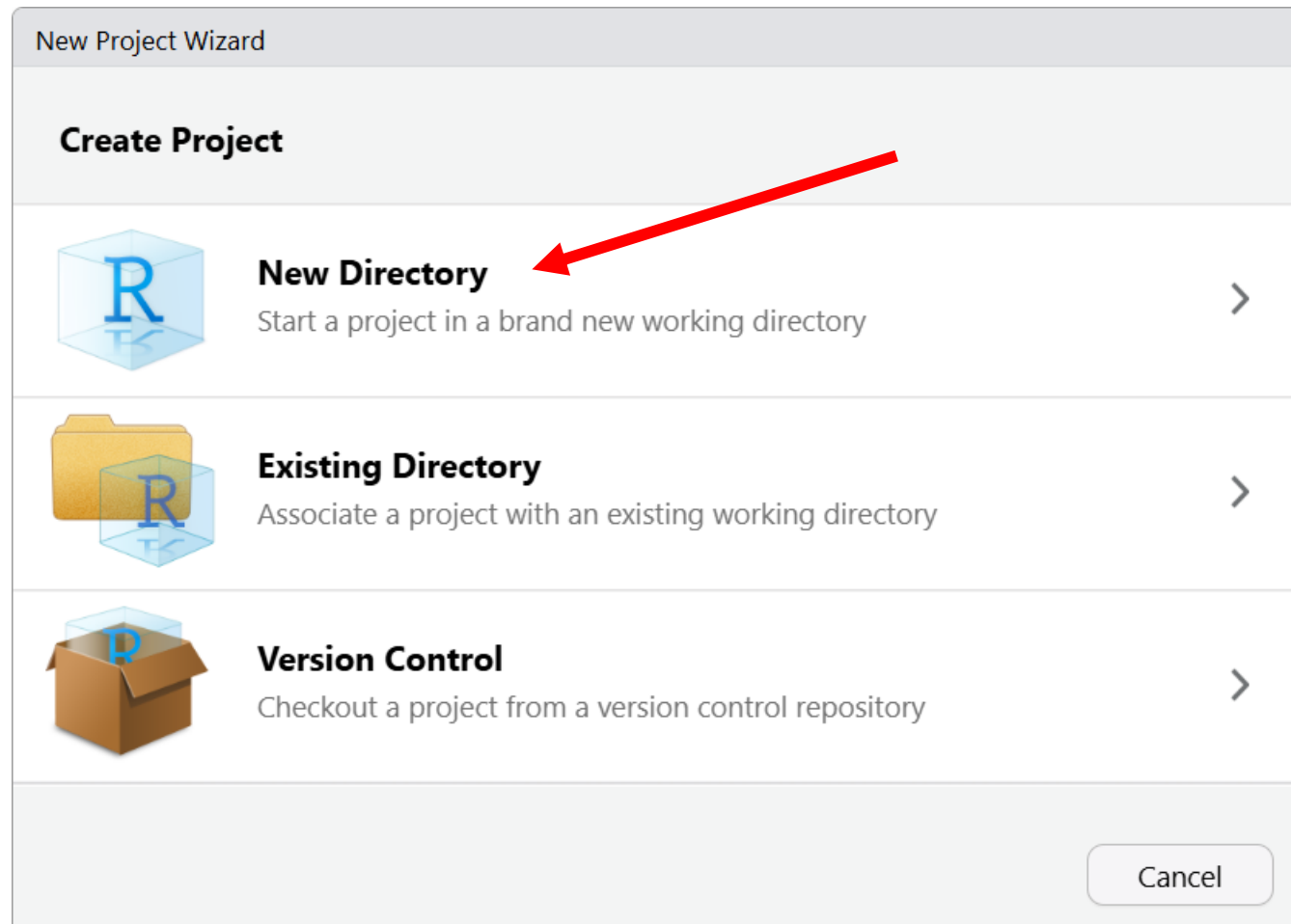
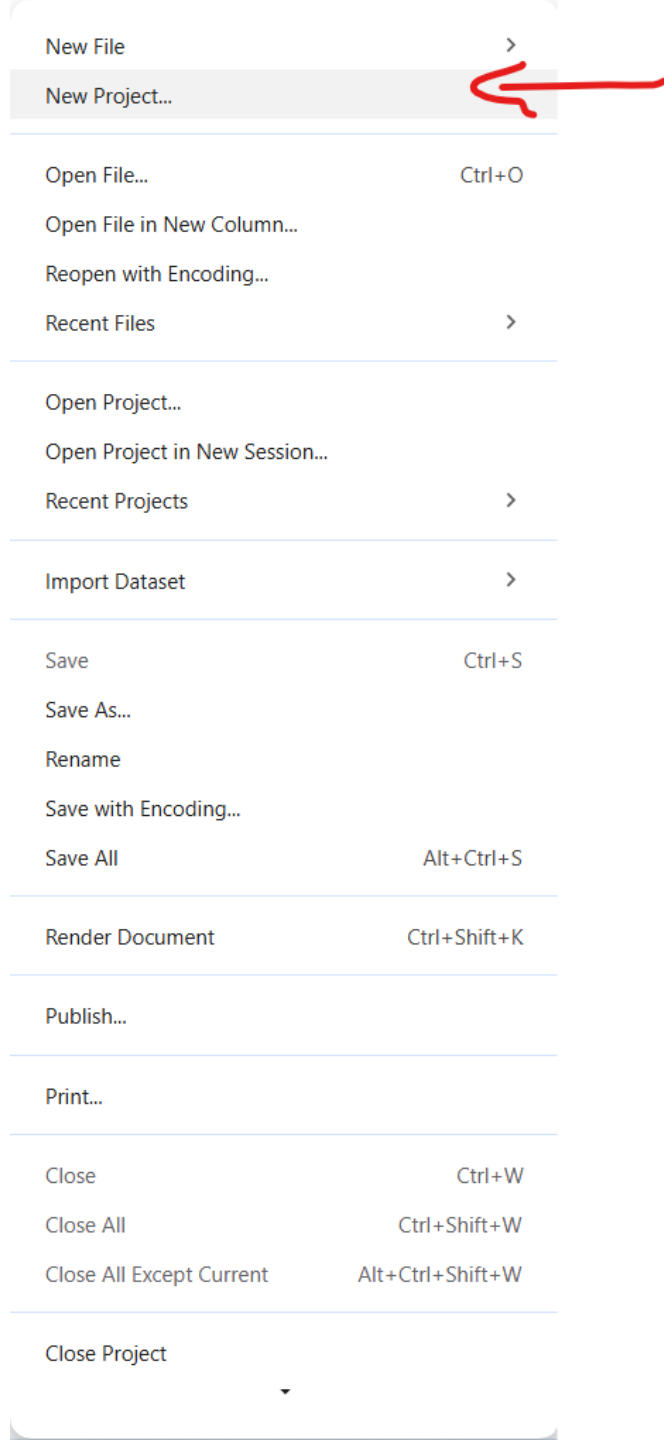
R

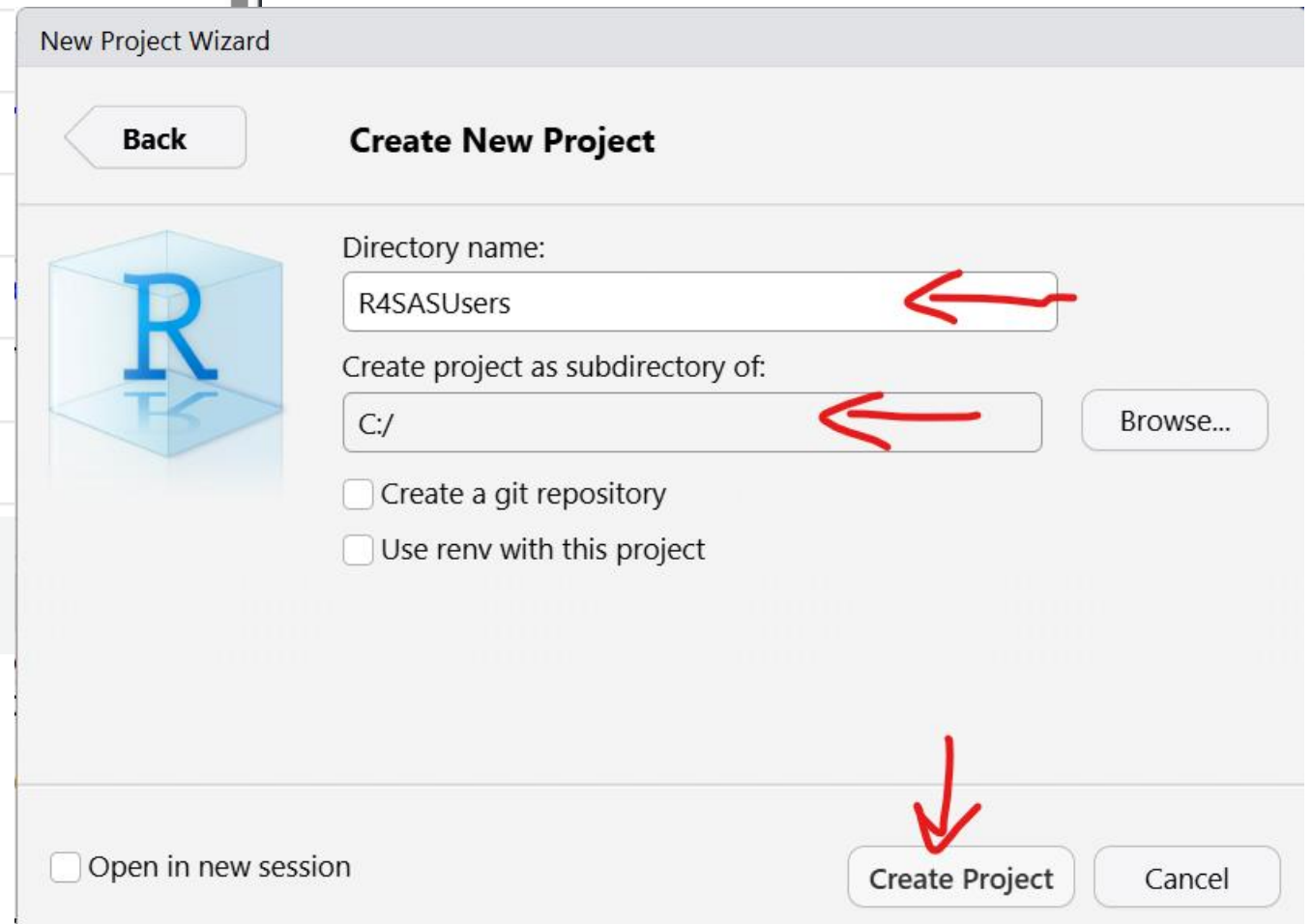
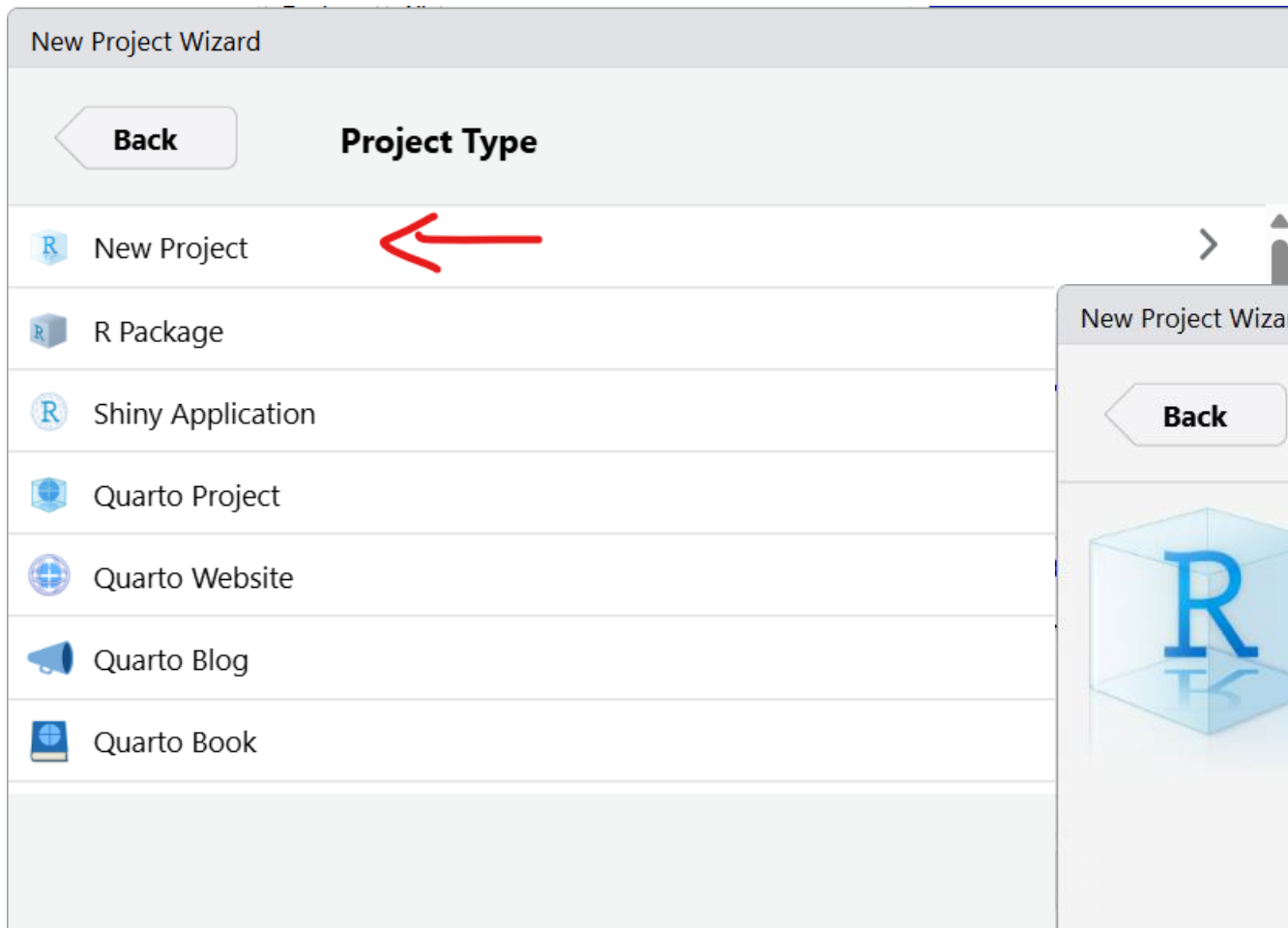
```
sessionInfo()  
library()
```

Open RStudio

Exercise 1:

- Look at session details
- Create new project “R4SASUsers”: File/New Project
- Put all files for R For SAS Users into this folder





Get Files From GoogleDrive Folder

My Drive > R4SASUsers_Sept2026



Type People Modified Source

74% of storage used If you run out, you can't create, edit, and upload files. Share 400 GB of storage with your family members for \$4.99/mo.



Clean up space

Get storage

Name	Owner	Date modified	File size	Sort
 FilesForRProject	 me	3:45 PM me	—	⋮
 OtherDatasets	 me	3:45 PM me	—	⋮
 RcodeFiles	 me	3:45 PM me	—	⋮
 Rdatasets	 me	3:45 PM me	—	⋮
 RmarkdownFiles	 me	3:45 PM me	—	⋮
 SAScodeFiles	 me	3:45 PM me	—	⋮
 SASdatasets	 me	3:45 PM me	—	⋮
 Slides	 me	3:45 PM me	—	⋮

Get Files From GoogleDrive Folder

My Drive > R4SASUsers_Sept2026 > FilesForRProject ▾



Type ▾

People ▾

Modified ▾

Source ▾

74% of storage used If you run out, you can't create, edit, and upload files. Share 400 GB of storage with your family members for \$4.99/mo.



[Clean up space](#)

[Get storage](#)

Name	Owner	Date modified	File size	Sort
CESDAnalysisReport_byRace.Rmd	me	3:42 PM me	6 KB	
CESDAnalysisReport_RaceParameters.Rmd	me	3:42 PM me	6 KB	
CESDAnalysisReport.Rmd	me	3:42 PM me	6 KB	
d1_homework_key.R	me	3:42 PM me	3 KB	
d1_homework.R	me	3:42 PM me	2 KB	
d2_homework_key.R	me	3:42 PM me	7 KB	
d2_homework.R	me	3:42 PM me	3 KB	
dat1.Rdata	me	3:42 PM me	2 KB	

Share (Ctrl+Alt+A)

R4SASUsers

New

Put Files Into This Folder

<

>

↑

This PC

OSDisk (C:)

R4SASUsers

Search R4SASUsers

Quick access

Emory

OneDrive - Emory University

This PC

Desktop

Documents

Downloads

Music

Pictures

Videos

OSDisk (C:)

Irclab (\\nasn2acts.cc.emory.edu\son-ts) (L

Projects (R:)

Dept (S:)

Personal (T:)

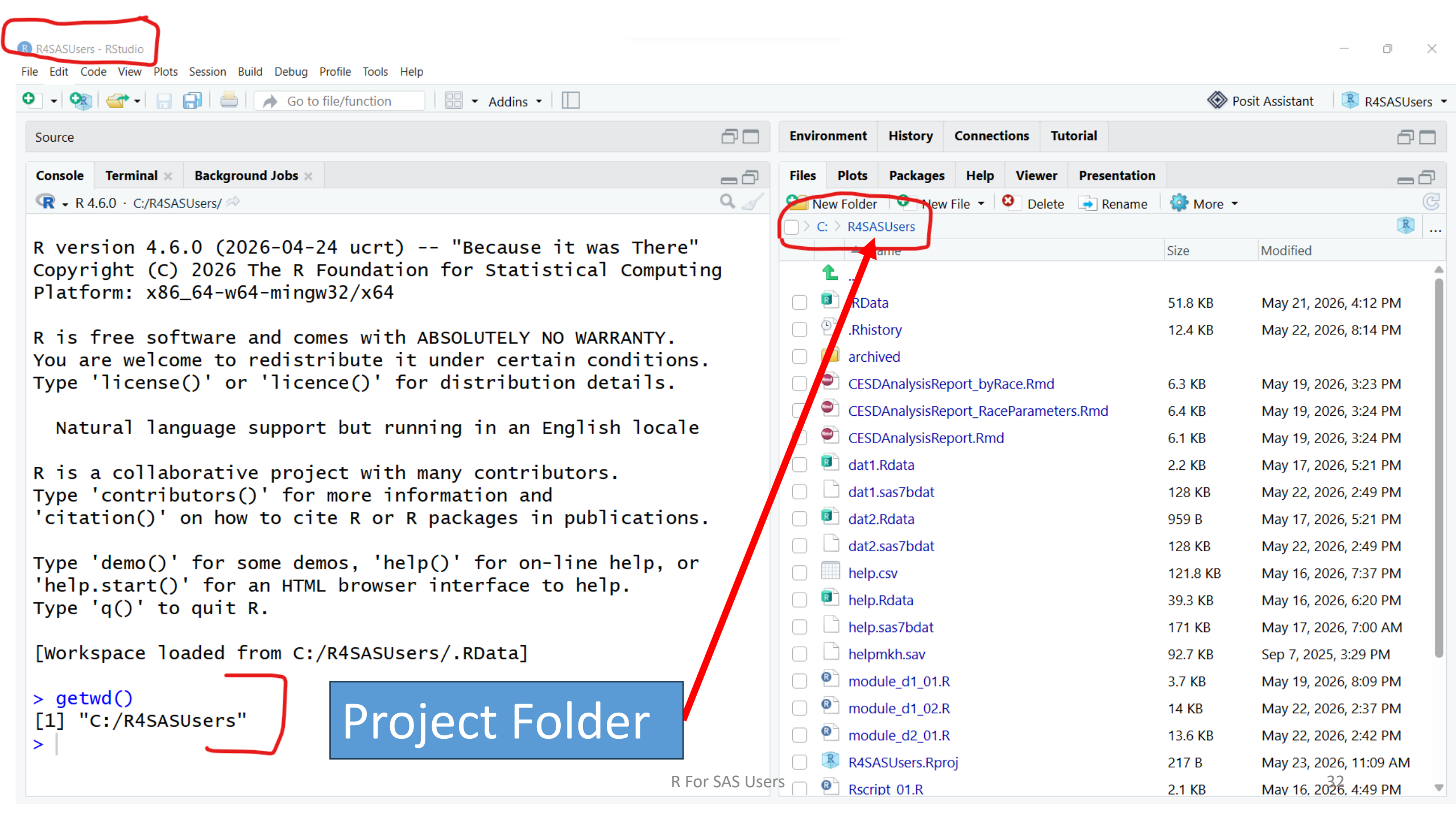
Network

Name	Date modified	Type	Size
.Rproj.user	5/10/2026 9:23 AM	File folder	
archived	5/19/2026 8:12 PM	File folder	
.RData	5/21/2026 4:12 PM	R Data File	
.Rhistory	5/22/2026 8:14 PM	R History Source File	
CESDAnalysisReport.Rmd	5/19/2026 3:24 PM	RMD File	
CESDAnalysisReport_byRace.Rmd	5/19/2026 3:23 PM	RMD File	
CESDAnalysisReport_RaceParameters.Rmd	5/19/2026 3:24 PM	RMD File	
dat1.Rdata	5/17/2026 5:21 PM	R Data File	
dat1.sas7bdat	5/22/2026 2:49 PM	SAS Data Set	1
dat2.Rdata	5/17/2026 5:21 PM	R Data File	
dat2.sas7bdat	5/22/2026 2:49 PM	SAS Data Set	1
help.csv	5/16/2026 7:37 PM	Microsoft Excel Comma Separ...	1
help.Rdata	5/16/2026 6:20 PM	R Data File	
help.sas7bdat	5/17/2026 7:00 AM	SAS Data Set	1
helpmkh.sav	9/7/2025 3:29 PM	SPSS Statistics Data Document	
module_d1_01.R	5/19/2026 8:09 PM	R File	
module_d1_02.R	5/22/2026 2:37 PM	R File	
module_d2_01.R	5/22/2026 2:42 PM	R File	

R For SAS Users

26 items

1 item selected 2.06 KB



R4SASUsers - RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function

Posit Assistant R4SASUsers

Source

Console Terminal Background Jobs

R 4.6.0 · C:/R4SASUsers/

R version 4.6.0 (2026-04-24 ucrt) -- "Because it was There"
Copyright (C) 2026 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

[Workspace loaded from C:/R4SASUsers/.RData]

```
> getwd()
[1] "C:/R4SASUsers"
>
```

Project Folder

Environment History Connections Tutorial

Files Plots Packages Help Viewer Presentation

New Folder New File Delete Rename More

> C: > R4SASUsers

	Name	Size	Modified
	RData	51.8 KB	May 21, 2026, 4:12 PM
	.Rhistory	12.4 KB	May 22, 2026, 8:14 PM
	archived		
	CESDAnalysisReport_byRace.Rmd	6.3 KB	May 19, 2026, 3:23 PM
	CESDAnalysisReport_RaceParameters.Rmd	6.4 KB	May 19, 2026, 3:24 PM
	CESDAnalysisReport.Rmd	6.1 KB	May 19, 2026, 3:24 PM
	dat1.Rdata	2.2 KB	May 17, 2026, 5:21 PM
	dat1.sas7bdat	128 KB	May 22, 2026, 2:49 PM
	dat2.Rdata	959 B	May 17, 2026, 5:21 PM
	dat2.sas7bdat	128 KB	May 22, 2026, 2:49 PM
	help.csv	121.8 KB	May 16, 2026, 7:37 PM
	help.Rdata	39.3 KB	May 16, 2026, 6:20 PM
	help.sas7bdat	171 KB	May 17, 2026, 7:00 AM
	helpmkh.sav	92.7 KB	Sep 7, 2025, 3:29 PM
	module_d1_01.R	3.7 KB	May 19, 2026, 8:09 PM
	module_d1_02.R	14 KB	May 22, 2026, 2:37 PM
	module_d2_01.R	13.6 KB	May 22, 2026, 2:42 PM
	R4SASUsers.Rproj	217 B	May 23, 2026, 11:09 AM
	Rscript 01.R	2.1 KB	May 16, 2026, 4:49 PM

R For SAS Users